

The Observable Dynamics Program: A Categorical Reading

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Abstract

This note develops the Observable Dynamics Program in categorical language, as a direct companion to the classical picture in *categorical_classical.tex*. The classical construction is algebra-first: a probability measure is freely chosen and marginals are derived from it. The program is diagram-first: the measure is derived from the coherence of a system of marginals. We trace this inversion through the same categorical layers — **Meas**, the Girmonad $\mathcal{P}$, its Kleisli category $\text{Kl}(\mathcal{P})$ of Markov kernels, and the extension theorem — and identify where the program’s arguments must go and what remains open. The inversion amounts to asking whether the forgetful functor from probability spaces to compatible marginal systems has a left adjoint, and under what conditions.

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1 The inversion: diagram-first probability

The classical construction of probability (see *categorical_classical.tex*, §6) has a single free parameter: the choice of measure $P \in \mathcal{P}X$ at Step 3. Everything else — pushforwards, marginals, kernels, projective limits — is forced by the categorical structure of **Meas** and $\text{Kl}(\mathcal{P})$.

The Observable Dynamics Program asks: can Step 3 be replaced by a derivation? Specifically: given a projective system of measurable spaces and a compatible family of probability measures on its finite-dimensional factors, is there a *canonical* probability measure on the projective limit that is forced by the coherence data, rather than freely chosen?

The program’s answer is yes — under conditions that we now identify categorically.

Classical (algebra-first)	Program (diagram-first)
Choose $P \in \mathcal{P}X$ freely	Derive P from cocone data
Marginals follow from P	P follows from marginals
Monad algebra assumed	Monad algebra forced
Step 3 is a free parameter	Step 3 is a theorem

2 Query systems as diagrams in $\mathbf{Kl}(\mathcal{P})$

Definition 1 (Query system). A query system $(\mathcal{Q}, \preceq)$ consists of:

- A directed preorder $(I, \leq)$ of query indices.
- For each $i \in I$, a measurable space $(O_i, \mathcal{O}_i)$ — the outcome space of query i .
- For each $i \leq j$, a measurable function $\pi_{ij} : O_j \rightarrow O_i$ — the refinement map from finer query j to coarser query i — satisfying $\pi_{ii} = \text{id}$ and $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ for $i \leq j \leq k$.

Remark 1 (Query systems as functors). A query system is a functor $F : I^{\text{op}} \rightarrow \mathbf{Meas}$, where I is the preorder category with objects the query indices and morphisms the order relations. This is precisely a projective system in $\mathbf{Meas}$.

The refinement maps π_{ij} are deterministic, so they embed into $\mathbf{Kl}(\mathcal{P})$ via the Dirac kernel $\hat{\pi}_{ij}(o, \cdot) = \delta_{\pi_{ij}(o)}$. The query system therefore lives in $\mathbf{Meas}$, but it is a diagram in $\mathbf{Kl}(\mathcal{P})$ under this embedding. We work in $\mathbf{Kl}(\mathcal{P})$ because the morphisms we ultimately care about — compatible marginals and extension kernels — are genuinely stochastic.

Definition 2 (Realization space). The realization space of a query system is the projective limit

$$\Omega = \varprojlim_{i \in I} O_i = \left\{ (o_i)_{i \in I} \in \prod_i O_i \mid \pi_{ij}(o_j) = o_i \text{ for all } i \leq j \right\}$$

equipped with the subspace σ -algebra from the product. The projections $\text{pr}_i : \Omega \rightarrow O_i$ are the canonical measurable maps.

Remark 2 (What Ω represents). A point $\omega \in \Omega$ is a coherent tuple of query outcomes: a choice of answer to every query simultaneously, consistent across all refinement relations. This is the program's analogue of a point in the sample space X — except Ω is constructed from the queries, not assumed.

In the delay query system (Paper 4 / observational_probability.tex), $\Omega \subseteq X^{\mathbb{Z}}$ is the set of bi-infinite sensor streams that are coherent under all delay-and-sample operations.

3 Compatible marginals as cocones

Definition 3 (Compatible marginals). A compatible family of probability measures for a query system $(\{O_i\}, \{\pi_{ij}\})$ is a collection $\{\nu_i \in \mathcal{P}O_i\}_{i \in I}$ satisfying

$$(\pi_{ij})_{\#} \nu_j = \nu_i \quad \text{for all } i \leq j.$$

Remark 3 (Cocone interpretation). In $\mathbf{Kl}(\mathcal{P})$, the compatible family $\{\nu_i\}$ is a cocone over the diagram $F : I^{\text{op}} \rightarrow \mathbf{Kl}(\mathcal{P})$. Specifically, each ν_i is a morphism $1 \rightarrow O_i$ in $\mathbf{Kl}(\mathcal{P})$ (a stochastic map from the terminal object to O_i), and the compatibility condition says that these morphisms commute with the refinement maps:

$$\begin{array}{ccc} 1 & \xrightarrow{\nu_j} & O_j \\ & \searrow \nu_i & \downarrow \hat{\pi}_{ij} \\ & & O_i \end{array}$$

The apex of the cocone is (if it exists) a measure P on Ω with $(\text{pr}_i)_\# P = \nu_i$ for all i .

The program’s central question is: does this cocone have a canonical apex?

Remark 4 (Classical vs. program direction). *Classically, the apex P is freely chosen first, and the cocone data $\{\nu_i\}$ are derived as pushforwards $(\text{pr}_i)_\# P$. This is a cone: P is the apex and the ν_i are its projections.*

The program runs in the opposite direction: the cocone data $\{\nu_i\}$ are given (from observations), and the apex P is derived. This is the categorical form of the epistemological inversion: the global measure is constituted by the coherence of its parts, not given as a prior that determines the parts.

4 The extension theorem as an adjunction

We now state the program’s main theorem — the Observational Extension Theorem of Paper 1 — in categorical language.

Definition 4 (Categories for the adjunction). *Define two categories:*

- **Prob**(Ω): *the category whose objects are probability measures $P \in \mathcal{P}\Omega$ and whose morphisms are measurable maps of probability spaces over Ω . (For route-finding purposes, we may take this to be a discrete category: objects are measures, no non-trivial morphisms.)*
- **Compat**($\{O_i\}$): *the category of compatible families $\{\nu_i\}$ over the query system, with morphisms being families of measure-preserving maps compatible with the refinements.*

Definition 5 (The forgetful functor). *The marginalisation functor*

$$\Phi : \mathbf{Prob}(\Omega) \rightarrow \mathbf{Compat}(\{O_i\})$$

sends a probability measure P on Ω to the compatible family of its marginals:

$$\Phi(P) = \{(\text{pr}_i)_\# P\}_{i \in I}.$$

This functor is always well-defined: pushforward is functorial and the compatibility condition holds automatically.

Theorem 1 (Extension theorem — adjunction form). *Under the conditions of Paper 1 (query system sequentially upper-directed and realizable), the marginalisation functor Φ is an isomorphism of categories: every compatible family $\{\nu_i\}$ extends uniquely to a measure P on Ω , and $\Phi(P) = \{\nu_i\}$. So Φ has a two-sided inverse (the extension functor) and the adjunction $\text{Ext} \dashv \Phi$ is in fact an equivalence.*

The careful development — the categories QSys, PSys, the functors, the unit/counit, and the proof (uniqueness = Paper 1 Determination Thm; existence = the Carathéodory route, realizability replacing tightness; Lean: `observational_extension` in `QuerySystem.lean`) — is in the companion `categorical_adjunction.tex` (Thms. on the adjunction and the equivalence). It is not repeated here.

Remark 5 (The Polish replacement). *The classical Kolmogorov extension theorem obtains existence by a topological route: Polish spaces are regular enough that Prokhorov’s tightness theorem guarantees inner regularity, which in turn forces σ -additivity.*

The program obtains existence by a purely measure-theoretic route: the realizability condition replaces inner regularity. Realizability says that the projections $\text{pr}_i : \Omega \rightarrow O_i$ are surjective, so that the projective limit is “full enough” that vanishing marginals imply vanishing cylinder content.

This is why the two approaches are complementary rather than redundant: the program’s route works in settings where the spaces need not be Polish (any standard measurable space with the right query structure), and the relationship to the Prokhorov/tightness route is precisely what the Conjecture 2 / CFA work on the `equicontinuity-cfa-conjecture` branch is probing.

5 The minimal predictive query in $\text{Kl}(\mathcal{P})$

We now identify the objects of Paper 2 as morphisms in $\text{Kl}(\mathcal{P})$.

Definition 6 (Delay query system). *Fix a measurable sensor alphabet $(X, \mathcal{X})$ and work on the path space $X^{\mathbb{Z}}$ with the product σ -algebra. The delay query system has:*

- Queries: pairs $(d, \tau) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$ with outcome space $O_{d,\tau} = X^d$ and evaluation map

$$\text{ev}_{d,\tau} : X^{\mathbb{Z}} \rightarrow X^d, \quad \omega \mapsto (\omega_0, \omega_{-\tau}, \omega_{-2\tau}, \dots, \omega_{-(d-1)\tau}).$$

- Refinement maps: $\pi_{(d,\tau),(d',\tau')} : X^{d'} \rightarrow X^d$ defined when $(d, \tau) \preceq (d', \tau')$, projecting to the corresponding coordinates.

The realization space is $\Omega \subseteq X^{\mathbb{Z}}$, the set of streams coherent under all delay evaluations.

Definition 7 (Predictive kernel as Kleisli morphism). *Given a compatible family $\{\nu_{d,\tau}\}$ and the extended measure P on Ω (from Theorem 1), the predictive kernel at time t is a morphism in $\text{Kl}(\mathcal{P})$:*

$$K_t : \Omega \rightarrow O_F, \quad K_t(\omega, \cdot) = P((\sigma^t \omega \in \cdot) \mid \mathcal{F}_{-\infty}^0),$$

where σ is the shift map and $O_F = X^{\mathbb{N}}$ is the future observable space. The minimal predictive query Q_* is the σ -algebra generated by $\{K_t\}_{t \geq 0}$ — the coarsest measurable structure on Ω from which all future conditional distributions can be computed.

Remark 6 (Semigroup in $\text{Kl}(\mathcal{P})$). *The Markov property of $\{K_t\}$ is exactly closure of this collection under Chapman–Kolmogorov composition in $\text{Kl}(\mathcal{P})$:*

$$K_{s+t} = K_t \circ K_s \quad (\text{equality in } \text{Kl}(\mathcal{P})).$$

The semigroup property of the predictive operator (Paper 2, Theorem 5.2) is associativity of composition in $\text{Kl}(\mathcal{P})$. The operator-theoretic content of Paper 3 (resolvent, spectral theory) is the analysis of this semigroup acting on $L^2(\Omega, P)$.

Remark 7 (The Rose condition as isomorphism). *Two delay embeddings (via different (d, τ) parameters) are observationally equivalent if their induced predictive sub-diagrams are isomorphic in $\text{Kl}(\mathcal{P})$. The Rose condition — $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$ — is the information-theoretic criterion for this isomorphism. It says that no f -divergence can distinguish the two predictive distributions, which in $\text{Kl}(\mathcal{P})$ means the two kernels are equal as morphisms.*

6 Open problems as categorical questions

The program’s open problems now have precise categorical formulations.

Conjecture 1 (Conjecture 2 — σ -additivity from coherence alone). *There exist purely algebraic conditions on the query system — without realizability or Polish hypotheses — that force the finitely additive content μ_0 to be σ -additive.*

Categorically: *what conditions on a diagram $F : I^{\text{op}} \rightarrow \text{Kl}(\mathcal{P})$ guarantee that the colimit exists in $\mathcal{P}\Omega$? This is a cocompleteness question for $\text{Kl}(\mathcal{P})$ restricted to diagrams of a particular shape.*

Connection to equicontinuity: *the CFA (Compact Family Argument) work on the **equicontinuity-cfa-con** branch is developing the Prokhorov-type tightness argument for this. If successful, it would give a topological route to cocompleteness analogous to the classical case, and potentially feed into a Mathlib formalization of the Kolmogorov extension theorem.*

Conjecture 2 (Stopping Point 1 — measurability from discriminability). *The σ -algebra $\mathcal{X}$ on the sensor alphabet X should be derivable from primitive discriminative structure: the ability of the observer to distinguish finite families of events.*

Categorically: *is there a category more primitive than **Meas** — perhaps of locales, domains, or topological spaces — whose image in **Meas** under a forgetful functor gives the right σ -algebras? The σ -algebra would then be the sheaf of reportable distinctions on a suitable site, and the assumption of $(X, \mathcal{X})$ in the current program would be replaced by a construction.*

Connection to Boole: *Boolean algebras of propositions (Boole’s algebra of thought) are the finite discriminative structure; the σ -algebra is their countable closure. The conjecture is that any system capable of coherent countable probabilistic reasoning must already have this closure.*

Remark 8 (The adjunction gap). *Theorem 1 states the extension theorem as an isomorphism of categories. The full adjunction statement — identifying the unit and counit, showing they satisfy the triangle identities, and characterising the fixpoints — has not been written out precisely. This is a concrete mathematical task: define the two categories properly, write down the functors Φ and Ext , and verify the adjunction axioms. It would give the cleanest categorical statement of what Paper 1 proves.*

7 Comparison table: classical vs. program

Step	Classical	Program
1. Starting point	Sample space $(X, \mathcal{X})$ assumed	Query system $(\{O_i\}, \{\pi_{ij}\})$ given
2. Ambient category	Meas and $\text{Kl}(\mathcal{P})$	Meas and $\text{Kl}(\mathcal{P})$ (same)
3. Measure	Free choice $P : 1 \rightarrow \mathcal{P}X$	Derived: $P = \text{Ext}(\{\nu_i\})$
4. Marginals	Derived from P	Given as input (observations)
5. Limit space	X is assumed	$\Omega = \varprojlim O_i$ is constructed
6. Extension	Kolmogorov (Polish)	Paper 1 (sequential, realizable)
7. Dynamics	Markov kernel $T : X \rightarrow X$ in $\text{Kl}(\mathcal{P})$	Predictive kernel $K_t : \Omega \rightarrow O_F$
8. Semigroup	Chapman–Kolmogorov	Composition in $\text{Kl}(\mathcal{P})$ (same)
9. Objectivity	Ground truth assumed	Limit of coherent query perspectives

Row 3 is the inversion. Rows 2 and 8 show that the categorical machinery is the same in both directions — the difference is not in the tools but in the direction of travel.

Row 9 deserves emphasis: the program does not eliminate objectivity but *derives* it. When all queries cohere — when the compatible family $\{\nu_i\}$ is globally consistent — the extension P is the unique probability measure on Ω that everyone with access to all queries would agree on. This is the mathematical form of the phenomenological claim that objectivity is constituted by the coherence of perspectives, not presupposed beneath them.

8 Mathlib availability for the program’s picture

See `mathlib_gaps.md` (this directory) for the consolidated audit (Kleisli / Eilenberg–Moore / Giry monad / Markov kernels present; projective limits and Kolmogorov extension absent; `QuerySystem.lean`’s `observational_extension` fills the extension gap in-house for the query setting). The one gap specific to *this* document: the adjunction of Theorem 1 (defining **Prob**(Ω) and $\text{Compat}(\{O_i\})$ as Lean 4 categories and verifying the adjunction axioms) is not formalized — the next categorical formalization target.